

STUDENT GRIEVANCE CELL
IT5613 SOCIALLY RELEVANT PROJECT LABORATORY

A PROJECT REPORT

Submitted by

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ABSTRACT

The student grievance cell system plays a crucial role in educational institutions, allowing students to voice their concerns and facilitating effective grievance resolution. This project focuses on developing a web application using React, Express.js, and MongoDB. The web application, developed, aims to streamline the grievance resolution process within the educational institution. Students can submit grievances, track their status, and communicate with authorized staff members. The application incorporates sentiment analysis to assess grievance urgency. Security features include authentication, authorization, route protection, password hashing, and the use of JSON Web Tokens (JWT). The project report covers system architecture, database design, implementation, and sentiment analysis integration. The system improves efficiency, identifies institutional problems, and enables anonymous grievance submission. It enhances student satisfaction and promotes a conducive learning environment.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF FIGURES	v
	LIST OF ABBREVIATIONS	vi
1.	INTRODUCTION	1
	1.1 Overview	1
	1.2 Objective	2
	1.3 Scope of the project	2
2.	LITERATURE SURVEY	4
	2.1 Analysis of existing solutions	4
3.	SYSTEM ARCHITECTURE AND DESIGN	6
	3.1 System Components	6
	3.2 System Architecture	7
	3.3 Database Design	7
	3.4 Authentication and Authorization	9
	3.5 System Communication and APIs	10
4.	IMPLEMENTATION OF THE PROJECT	11
	4.1 Technology Stack	11
	4.2 Front-End Implementation	11
	4.3 Back-End Implementation	13
	4.4 Database Integration	15
	4.5 Third Party Services	16
5.	CONCLUSION AND FUTURE WORK	18
	5.1 Conclusion	18
	5.2 Future Work	19
	REFERENCES	20

LIST OF FIGURES

FIGURE. NO.	TITLE	PAGE NO.
3.1	Architecture diagram	6
3.2	A sample grievance model stored in MongoDB	9
3.3	JWT Authentication Scheme	10
4.1	MERN Stack Architecture	11
4.2	Homepage of the application	12
4.3	The grievance details page of a sample grievance	13
4.4	Page listing the grievances assigned to a staff	13
4.5	JWT token stored in a cookie in the browser	15
4.6	List of collections stored in MongoDB	16
4.7	Server log showing the successful verification of a CAPTCHA	17
4.8	Sentiment of a grievance fetched from Hugging Face API	17

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
CAPTCHA	Completely Automated Public Turing test to tell Computers and Humans Apart
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express, React, Node
NLP	Natural Language Processing
NoSQL	Not only Structured Query Language
ODM	Object Data Modeling
ORM	Object Relational Mapper
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The student grievance cell is a vital component of any educational institution, providing a platform for students to voice their concerns, seek resolution, and contribute to the overall improvement of the institution. In this project, we have developed a web-based student grievance cell system to streamline the grievance resolution process and enhance student satisfaction.

Our system offers an intuitive and user-friendly interface that allows students to easily raise grievances, track the status of their submissions, and engage in effective communication with authorized staff members. Through the implementation of features such as user authentication, authorization, and protection of routes, we ensure the security and privacy of the system.

To improve the efficiency of the grievance resolution process, our system incorporates sentiment analysis. This analysis helps in identifying the urgency and severity of grievances, enabling staff members to prioritize and address them accordingly.

In addition, we have integrated third-party services like reCAPTCHA to prevent automated bot submissions and maintain the integrity of the system. The use of reCAPTCHA ensures that only genuine human users can submit grievances, minimizing the risk of spam or malicious activity.

Our system also allows for anonymous grievance submission, respecting the privacy and confidentiality of students who may prefer not to disclose their identities. This feature encourages students to freely express their concerns without any fear of reprisal.

Furthermore, the system provides administrators and authorized staff members with comprehensive tools for managing and resolving grievances. It includes features such as dashboard analytics, tracking of grievance history, and reporting functionalities to identify recurring issues and track the performance of the grievance resolution process.

Overall, our student grievance cell system aims to provide a transparent and efficient platform for students to raise and address grievances. By facilitating effective communication, prioritizing urgent grievances, and maintaining security and privacy, we strive to enhance student satisfaction and contribute to a positive learning environment within the institution.

1.2 OBJECTIVE

The objective of the student grievance cell project is to create an efficient and transparent platform for students to raise and track grievances within an educational institution. The project aims to achieve the following objectives:

- **Streamline Grievance Resolution:** Provide a centralized system where students can easily submit their grievances, ensuring that their concerns are heard and addressed promptly.
- **Track Grievance Status:** Enable students to track the progress and status of their grievances, promoting transparency and accountability in the resolution process.
- **Enhance Efficiency:** Automate and streamline the grievance management process, reducing manual efforts and paperwork. This improves the efficiency of handling grievances and enables faster resolution.
- **Ensure Security and Privacy:** Implement robust authentication and authorization mechanisms to ensure that only authorized users can access and interact with the system. Also, provide the option for anonymous grievance submission to protect the privacy of students who prefer not to disclose their identities.
- **Improve Student Satisfaction:** By providing a user-friendly and responsive platform for grievance management, the project aims to enhance student satisfaction.

1.3 SCOPE OF THE PROJECT

The scope of the student grievance cell project encompasses the development of a web application that allows students to raise and track grievances within an educational institution. The project focuses on providing a user-friendly interface for students to

submit their grievances, track the status of their submissions, and communicate with authorized staff members. The system incorporates sentiment analysis to assess grievance urgency and prioritize resolutions. It includes features such as user authentication, authorization, protection of routes, and integration with third-party services like reCAPTCHA for security and the Hugging Face API for sentiment analysis. The project's scope also includes database integration using MongoDB with Mongoose as the ORM. The goal is to enhance the efficiency of the grievance resolution process, identify institutional problems, ensure security and privacy, and ultimately improve student satisfaction within the educational institution.

CHAPTER 2

LITERATURE SURVEY

2.1 ANALYSIS OF EXISTING SOLUTIONS

Numerous studies and research papers have focused on grievance redressal systems across various domains, including government services, consumer affairs, and organizational management. Existing solutions typically involve manual handling of complaints through traditional mediums like letters or email.

Traditional Letters:

Historically, grievance redressal systems have relied on written letters to register and address complaints. The process of handling and responding to letters requires significant manual effort, leading to slower resolution times.

Email-Based Systems:

With the advent of email, many education institutions have transitioned to email-based grievance redressal systems. While this method offers faster communication compared to letters, it still lacks the centralized platform necessary for streamlined and efficient complaint management. Email systems often suffer from issues such as scattered correspondence, lack of automated tracking, and difficulty in aggregating and analyzing data.

Despite the extensive research on grievance redressal systems, there is a notable gap in the literature regarding centralized platforms for efficient complaint handling. The majority of existing solutions focus on the traditional methods of communication, such as letters or email, which lack the advantages provided by centralized platforms.

Opportunities for Centralized Platforms:

Centralized platforms offer several advantages over traditional grievance redressal methods. By utilizing a centralized platform, organizations can achieve:

Streamlined Complaint Management: A centralized platform allows for the consolidation and organization of complaints, ensuring that all grievances are systematically recorded, tracked, and addressed.

Efficient Communication: With a centralized platform, communication between complainants and the redressal authority becomes more efficient and transparent. Instant notifications, updates, and two-way communication channels can significantly improve the overall user experience.

Data Analysis and Decision Support: Centralized platforms enable the collection and analysis of grievance data, providing valuable insights for decision-making, identifying trends, and implementing proactive measures to prevent recurring issues.

Accountability and Monitoring: A centralized platform allows for better monitoring of the grievance redressal process, ensuring accountability and transparency at each stage. It facilitates the tracking of complaint resolution timelines and the performance evaluation of redressal authorities.

Anonymous Grievance Raising Technology: The incorporation of anonymous grievance reporting system in the student grievance cell will enable students to express their concerns without fear of reprisal and offers several advantages like protection of identity, enhanced participation, early intervention, deterrent for misconduct, and so on.

Sentiment Analysis: It enables the identification of urgency of the grievances raised, allowing for timely intervention, improved response time and data-driven decision-making.

CHAPTER 3

SYSTEM ARCHITECTURE & DESIGN

3.1 OVERVIEW

The student grievance cell system is designed to provide a streamlined and efficient platform for students to raise grievances and for authorities to manage and resolve them effectively. The system architecture is built using React for the front-end, Express.js for the back-end, and MongoDB for the database.

Security is a crucial aspect of the system architecture. The system incorporates several security measures, including the use of reCAPTCHA to verify users and prevent bot submissions. Password hashing is implemented to protect user passwords, ensuring they are stored securely. Protected routes are utilized to restrict access to authorized users only, preventing unauthorized access to sensitive data.

Overall, the system architecture and design focus on providing a secure, user-friendly, and efficient platform for students to raise grievances and for authorities to manage and resolve them.

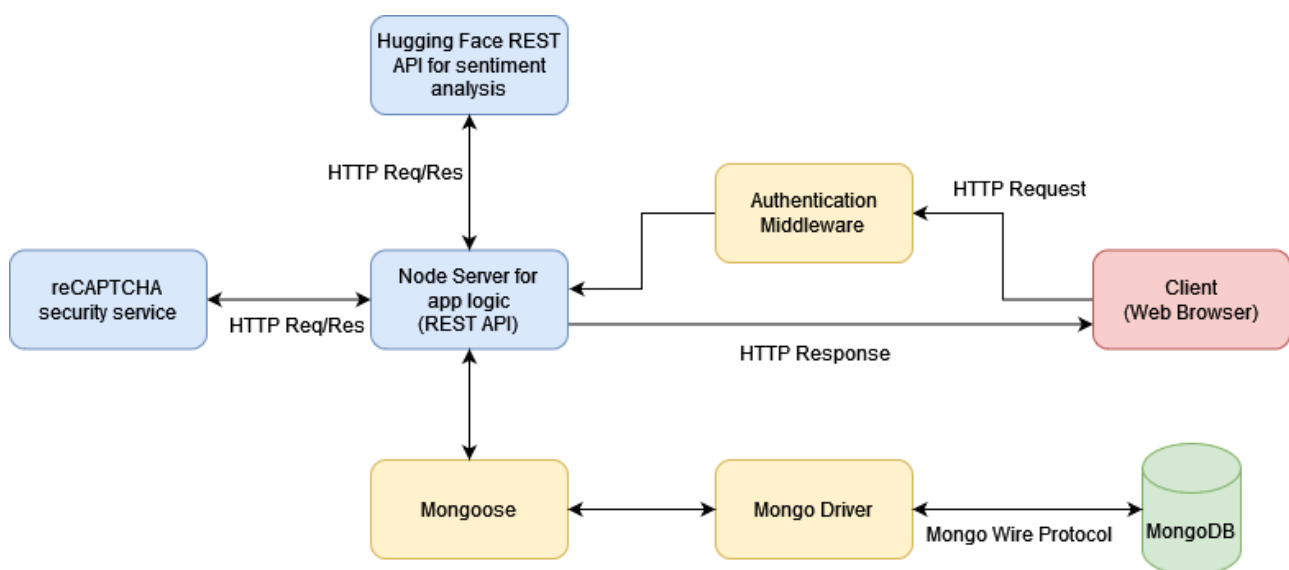


Figure 3.1 Architecture diagram

3.2 SYSTEM COMPONENTS

The student grievance cell system consists of several components that work together to provide a comprehensive platform for managing grievances. These components include:

User Interface (UI): The user interface component is responsible for presenting the system's features and functionalities to the users. It encompasses the design and layout of the web application, allowing students to submit grievances, track their status, and access relevant information.

Backend Server: The backend server component is built using Express.js, a flexible and scalable Node.js framework. It handles incoming requests from the frontend, manages routing, and performs the necessary business logic.

Database: The database component stores and manages the grievance data. MongoDB, a NoSQL document database, is commonly used due to its flexibility and scalability.

Authentication and Authorization: The authentication and authorization component ensures that only authenticated and authorized users can access the system. It handles user registration, login, and session management.

Third-Party Integrations: The system also incorporates various third-party integrations to enhance its functionality. For example, it integrates with reCAPTCHA to verify user submissions and prevent automated bot submissions. Additionally, email notifications can be integrated to inform users about the status updates of their grievances, ensuring effective communication and transparency.

3.3 DATABASE DESIGN

The student grievance cell system utilizes MongoDB as the database, and the database design is implemented using Mongoose, an Object-Relational Mapping (ORM) library for MongoDB. The design includes several models to represent different entities and their relationships within the system. The following models have been created to organize and manage the data effectively:

1. Admin: The Admin model represents the administrative users of the system. It stores information such as the admin's username, email, and hashed password.
2. Anonymous Grievance: The Anonymous Grievance model represents the grievances submitted anonymously by students. It includes fields such as the grievance description, timestamp, and sentiment analysis results (if applicable).
3. Comment: The Comment model is used to store comments made by authorities or staff members in response to a grievance. It includes fields such as the comment text, timestamp, and the user who made the comment.
4. Department: The Department model represents the different departments. It includes fields such as the department name and any additional information related to the department.
5. Grievance: The Grievance model represents the grievances raised by students. It includes fields such as the student's information, grievance description, timestamp, and the status of the grievance.
6. Grievance Status: The Grievance Status model represents the different stages or statuses of a grievance, such as "Pending," "In Progress," or "Resolved." It stores the status name and any additional information related to the status.
7. Grievance Type: The Grievance Type model represents the different types or categories of grievances that students can raise. It includes fields such as the type name and any additional information related to the grievance type.
8. Staff: The Staff model represents the authorized staff members or authorities who are responsible for managing and resolving grievances. It stores information such as the staff member's name, designation, and login credentials.
9. Student: The Student model represents the student users of the system. It includes fields such as the student's name, email, and login credentials.

These models define the structure and relationships between various entities in the database. They enable efficient storage, retrieval, and querying of the grievance-related

data, supporting the core functionalities of the student grievance cell system. The use of Mongoose as the ORM simplifies the interaction with the MongoDB database, providing a convenient and consistent interface for accessing and manipulating the data.

```
_id: ObjectId('646f67d4bba405c2af374550')
title: "Neglected Infrastructure: Broken Tiles in the Classroom"
description: "I express my deep concern about the persistently broken tiles in the c..."
student: ObjectId('646f66ce4c5c61daea13d1c2')
grievanceType: ObjectId('641db82a417f9d2e46606b0e')
grievanceStatus: ObjectId('641db5d9225e4edcf045270c')
staffAssigned: ObjectId('646f655bab811752844f7fd9')
▶ comments: Array
createdAt: 2023-05-25T13:51:16.637+00:00
updatedAt: 2023-05-25T13:52:52.483+00:00
__v: 0
```

Figure 3.2 A sample grievance model stored in MongoDB

3.4 AUTHENTICATION AND AUTHORIZATION

The student grievance cell system incorporates authentication and authorization for secure access control. Users are authenticated by validating their credentials during registration and login. Password hashing techniques ensure password security.

Routes are protected using middleware that checks user authentication and authorization before granting access. Admins, staff, and students have different levels of access based on their roles and privileges.

JWT (JSON Web Tokens): JSON Web Tokens (JWT) are used for secure and stateless authentication in the system. When a user logs in, the system generates a JWT containing encrypted user information and signs it with a secret key. This token is then sent to the client and stored locally. With each subsequent request to protected routes, the client includes the JWT in the request headers. The server verifies the integrity of the token and extracts the user information to perform authentication and authorization checks.

By utilizing JWT, the system eliminates the need for session management and enables stateless authentication. It provides a secure and efficient method for validating the

authenticity and permissions of users without relying on server-side session storage.

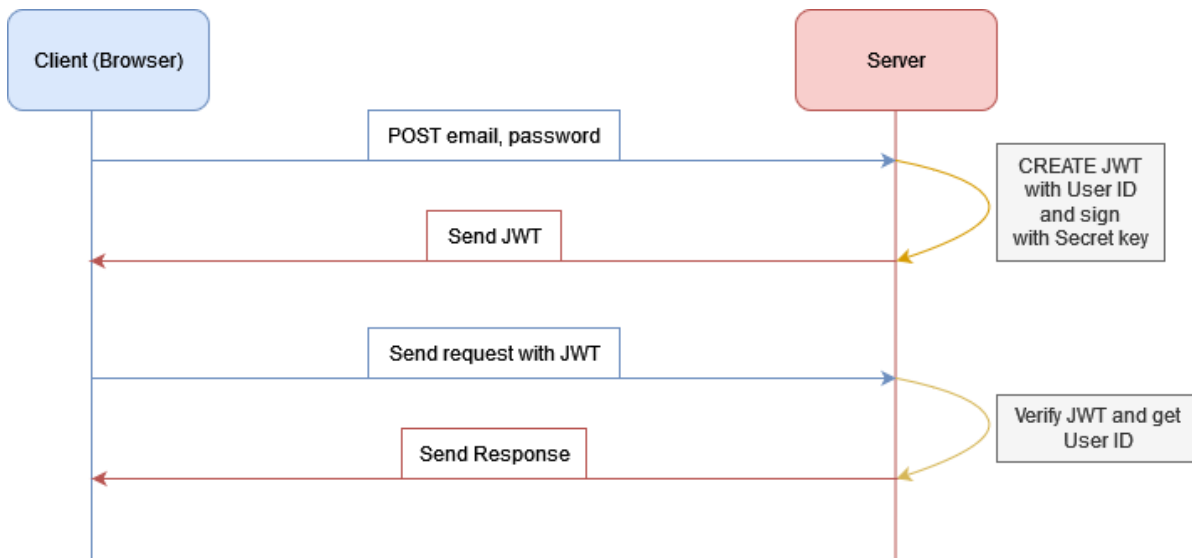


Figure 3.3 JWT Authentication Scheme

3.5 SYSTEM COMMUNICATION AND APIs

The student grievance cell system facilitates communication between different components through REST APIs (Representational State Transfer).

For example, to submit a grievance, the front-end sends an HTTP POST request to the corresponding API endpoint, which triggers the back-end to process the request and store the grievance in the database.

The communication between the front-end and back-end is achieved through exchanging JSON (JavaScript Object Notation) data. JSON provides a lightweight and readable format for data transmission and is widely supported by both front-end and back-end technologies.

CHAPTER 4

IMPLEMENTATION OF THE PROPOSED WORK

4.1 TECHNOLOGY STACK

The student grievance cell project is implemented using the MERN stack, which stands for MongoDB, Express.js, React, and Node.js. This powerful and popular technology stack allows for efficient development of a full-stack web application. The MERN stack empowers the student grievance cell project with a comprehensive and integrated set of technologies. It allows for seamless communication between the front-end and back-end components, ensuring a smooth user experience. The stack's flexibility, scalability, and performance benefits make it an ideal choice for developing a robust and feature-rich web application.

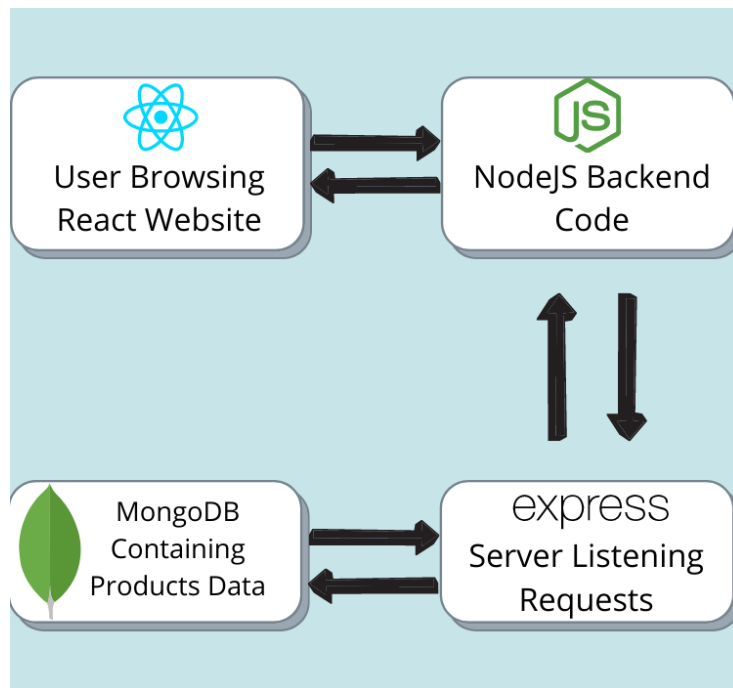


Figure 4.1 MERN Stack Architecture

4.2 FRONT-END IMPLEMENTATION

The front-end implementation of the student grievance cell project utilizes React, a powerful JavaScript library for building user interfaces. React enables the creation of

a dynamic and interactive user interface that enhances the user experience.

Components: The project is organized into reusable components, which are modular building blocks of the user interface.

State Management: React's state management capabilities, including hooks and context API, are leveraged to manage the application's state.

Responsive Design: The front-end implementation incorporates responsive design principles, ensuring that the application is accessible and optimized for various devices and screen sizes.

API Integration: The front-end interacts with the back-end through RESTful APIs. The Axios library is utilized to make HTTP requests to the server, retrieving and updating data from the database. This enables real-time data exchange between the front-end and back-end components of the application.

Overall, the front-end implementation of the student grievance cell project focuses on creating an intuitive and seamless user experience.

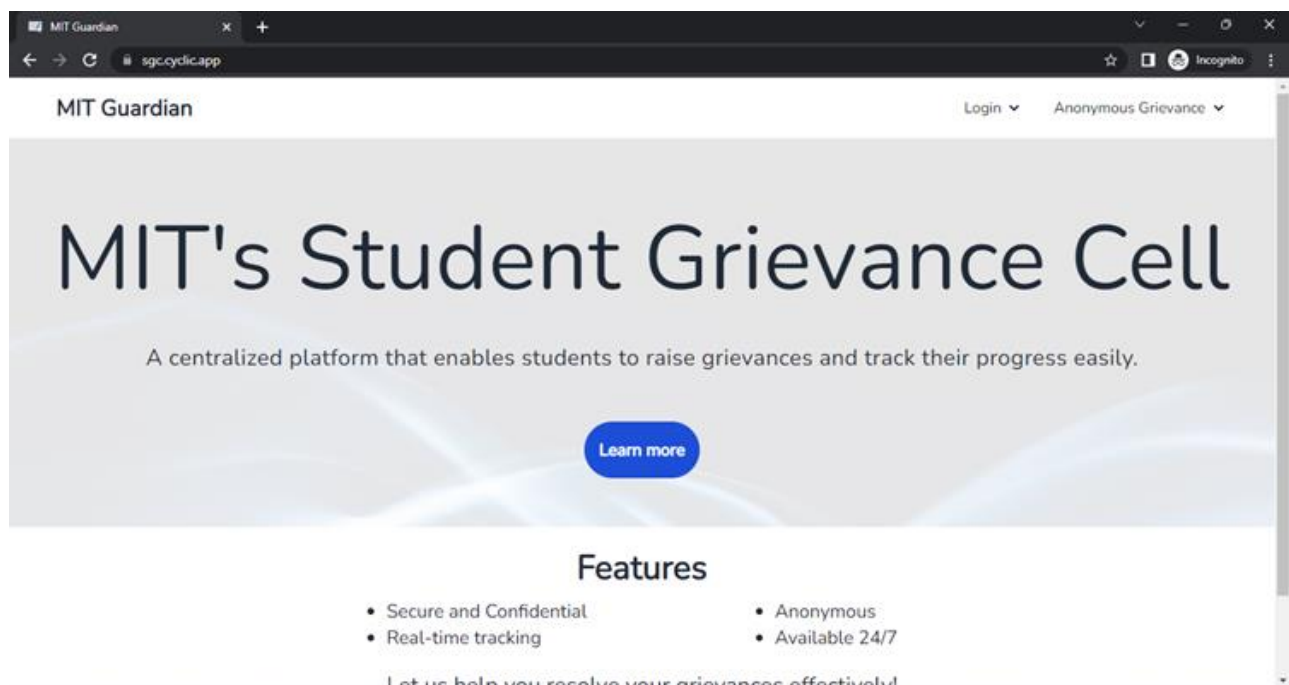


Figure 4.2 Homepage of the application

Inadequate Connectivity: Lack of WiFi in the Campus

Anonymous grievance

Created: 7:25 pm, 25/05/2023

Tracking ID: qXn51AF3

Description: We, the students, are frustrated with the lack of WiFi connectivity on campus. It hinders our access to online resources and communication, adversely affecting our learning experience. Urgent action is requested to address this issue and improve connectivity.

Staff Assigned: Raj

Status: Resolved ⓘ

Category: Infrastructure

Comments

No comments yet

Add any information you think is relevant

ⓘ Your comments will be anonymous and visible to the admin and the staff assigned to your grievance

Figure 4.3 The grievance details page of a sample grievance

Grievances assigned to you

Filters

Grievance Status: All Grievance Type: All

Regular Grievances

Neglected Infrastructure: Broken Tiles in the Classroom

Pragadesh BIS (2020506064)

Type: Infrastructure | Status: In Progress

Last updated: 7:22 pm, 25 May 23

I express my deep concern about the persistently broken tiles in the classroom. The safety of studen...

Items Per page 6

< previous 1 next >

Anonymous Grievances

Inadequate Connectivity: Lack of WiFi in the Campus

Anonymous

Type: Infrastructure | Status: Resolved

Last updated: 7:25 pm, 25 May 23

We, the students, are frustrated with the lack of WiFi connectivity on campus. It hinders

Figure 4.4 Page listing the grievances assigned to a staff

4.3 BACK-END IMPLEMENTATION

The backend implementation of the student grievance cell project is built using Express.js and Node.js, providing a robust and scalable server-side architecture. It handles the processing of requests, data storage and retrieval, authentication, and authorization.

Server Configuration: The Node.js runtime environment is set up with Express.js, a

minimalist web application framework. The server is configured to listen for incoming requests on a specified port and handle them accordingly.

Routing and API Endpoints: Express.js facilitates the creation of API endpoints that define the server's routes and handle the corresponding requests. These endpoints handle actions such as submitting grievances, fetching grievance data, managing user authentication and authorization, and handling other system functionalities.

Database Integration: The backend integrates with MongoDB, a NoSQL document database, using the Mongoose library. Mongoose provides an object data modeling (ODM) approach, allowing developers to define schemas and interact with the database in a structured manner.

Authentication and Authorization: The backend implements authentication and authorization mechanisms to ensure secure access to the system. User registration and login functionalities are implemented, validating user credentials and generating JSON Web Tokens (JWT) upon successful authentication.

Middleware: Express.js middleware is employed to perform various tasks, such as parsing request bodies, logging, and error handling.

Error Handling: The backend is equipped with error handling mechanisms to gracefully handle and respond to any errors that may occur during the request processing. Error messages are properly formatted and returned to the client, providing meaningful feedback.

The backend implementation focuses on providing secure and efficient server-side functionalities. Express.js and Node.js enable the creation of a scalable and performant server.

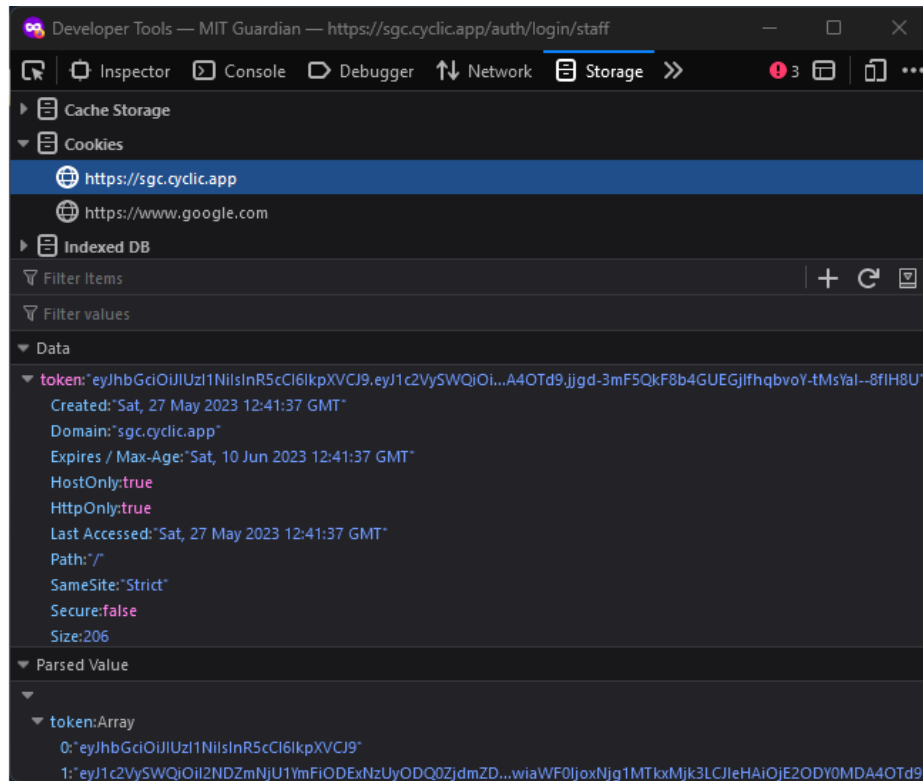


Figure 4.5 JWT token stored in a cookie in the browser

4.4 DATABASE INTEGRATION

The student grievance cell project integrates with MongoDB, a powerful NoSQL document database, using the Mongoose library. The database integration plays a crucial role in storing and retrieving grievance-related data efficiently and securely. Mongoose provides an Object Data Modeling (ODM) approach that allows developers to define schemas for the various entities in the system, such as grievances, comments, users, departments, and grievance types. Schemas define the structure, data types, and relationships between the entities, ensuring data consistency and integrity.

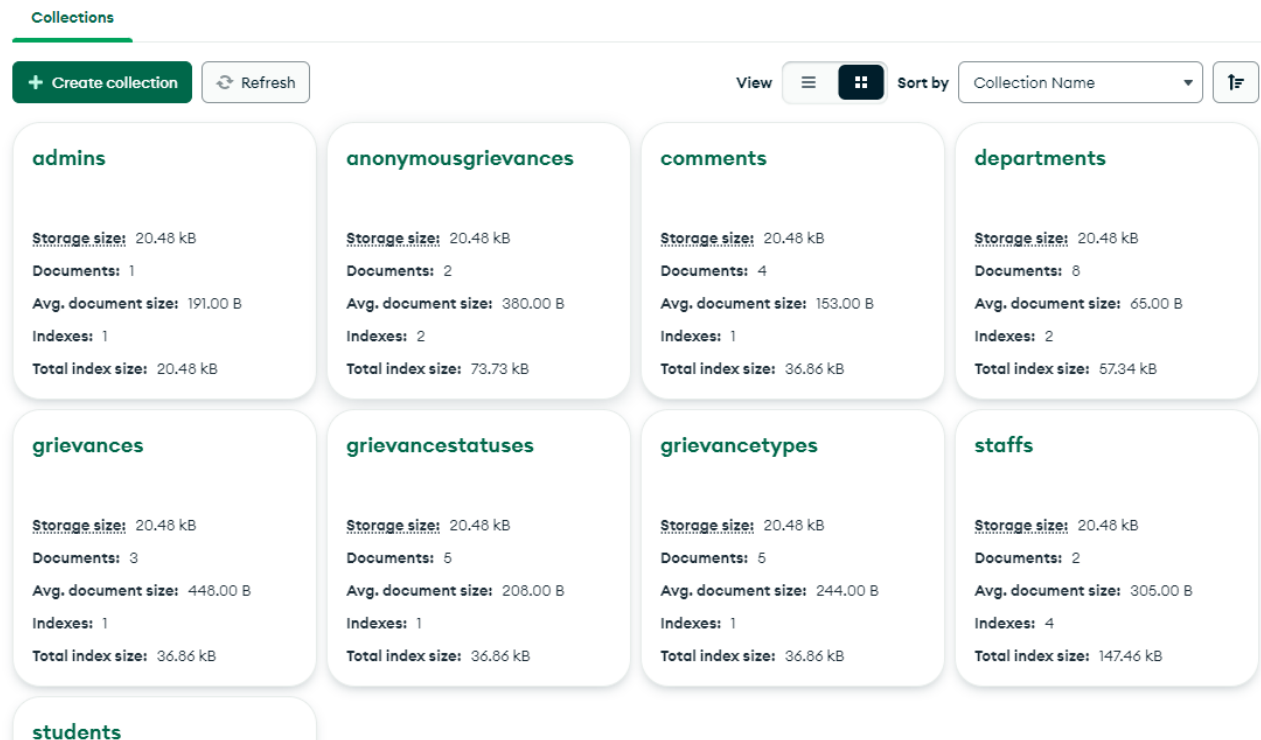


Figure 4.6 List of collections stored in MongoDB

4.5 THIRD-PARTY SERVICES

The student grievance cell project incorporates the integration of third-party services to enhance its functionality and provide additional features.

reCAPTCHA Integration: To enhance security and prevent automated bot submissions, the project integrates reCAPTCHA. reCAPTCHA is a widely-used service provided by Google that distinguishes between human and automated bot traffic. By integrating reCAPTCHA, the project ensures that only genuine human users can submit grievances and interact with the system, mitigating the risk of spam or malicious activity.

Hugging Face API for Sentiment Analysis: The project leverages the Hugging Face API, a popular natural language processing (NLP) service, for sentiment analysis. By integrating the Hugging Face API, the project gains access to state-of-the-art NLP models that can accurately analyze the sentiment of textual data. This analysis helps prioritize and address grievances based on their urgency.

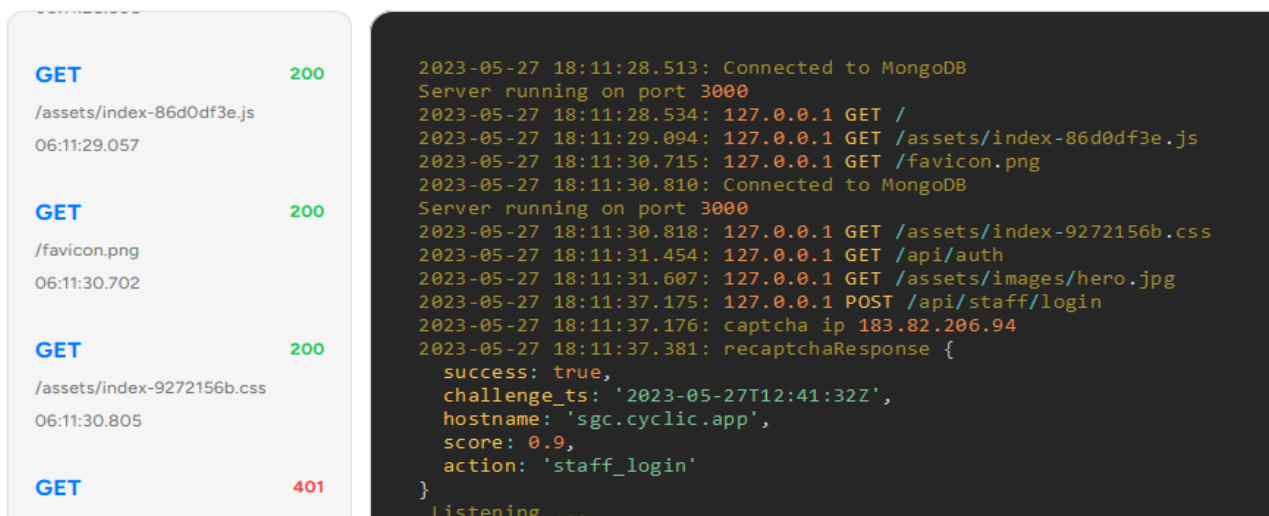


Figure 4.7 Server log showing the successful verification of a CAPTCHA

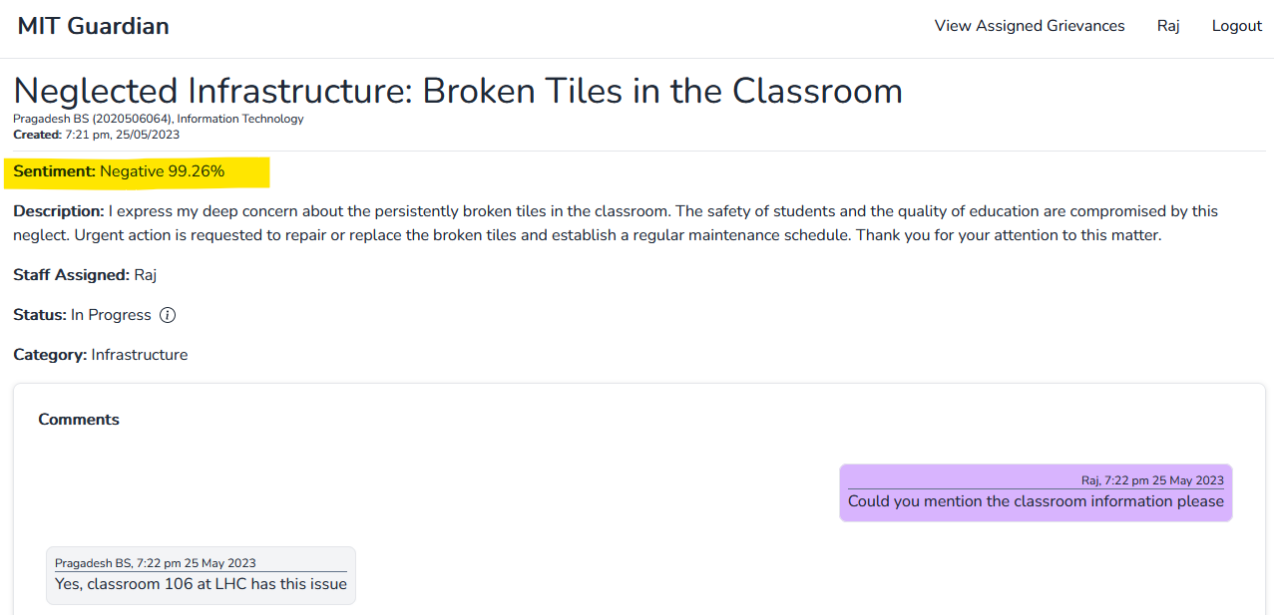


Figure 4.8 Sentiment of a grievance fetched from Hugging Face API

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION OF THE WORK

The development and implementation of the project, a student grievance cell, has been a significant milestone in addressing the concerns and issues faced by students within our institution. Through the careful integration of various features and functionalities, the platform has successfully provided a centralized, secure, and confidential space for students to submit and track their grievances, while allowing authorities to manage and address them efficiently. One of the key achievements of the project is the provision of anonymous grievance submissions. By allowing students to express their concerns without revealing their identities, we have created a safe environment where they can freely communicate their grievances without fear of repercussions. This feature has been crucial in encouraging students who may be hesitant to come forward due to concerns about confidentiality or potential consequences.

The categorization of grievances into different categories, such as Academics, Infrastructure, Library, and Administration, has enabled streamlined management and resolution of issues. This categorization allows for the appropriate allocation of grievances to the relevant departments or staff members, ensuring that they are addressed by those with the necessary expertise and authority.

The tracking functionality provided to students has been instrumental in promoting transparency and accountability. By enabling students to monitor the progress of their grievances, they can have a clear understanding of the actions being taken and the expected timeline for resolution. The communication and collaboration aspect of the platform, facilitated through the commenting and chatting facility between assigned staff and students, has enhanced the efficiency of grievance resolution.

In conclusion, the Student Grievance Cell Project has successfully addressed the needs and concerns of our student community by providing an efficient, secure, and confidential grievance resolution platform.

5.2 FUTURE WORK

The future enhancements for the student grievance system encompass various aspects to further optimize its functionality and user experience. One crucial addition would be the integration of intelligent chatbots as virtual assistants within the system. These chatbots would be equipped with natural language understanding capabilities, allowing them to engage in interactive conversations with students. The chatbots can assist students in submitting grievances, provide basic guidance and support, and address frequently asked questions, enhancing the overall user experience.

Automation can play a crucial role in improving the efficiency of the grievance resolution process. Implementing automated reminders for pending grievances and incorporating escalation protocols for unresolved issues would ensure timely attention and prompt resolution. Furthermore, escalation mechanisms would automatically raise unresolved issues to higher authorities, ensuring that critical grievances receive the appropriate attention and are resolved within the specified timelines.

Considering the increasing usage of mobile devices, developing a dedicated mobile application for the student grievance system would be a significant step forward. The user-friendly interface optimized for mobile screens would ensure seamless engagement and communication, allowing students to conveniently participate in the grievance resolution process.

To foster a sense of community and support, a peer mentoring program could be integrated into the student grievance system. This feature would enable experienced students or alumni to volunteer as mentors, guiding their peers through the grievance resolution process. Peer mentors would provide practical advice, emotional support, and share their personal experiences, creating a supportive network within the institution and helping students navigate the system effectively.

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